

For the use of a Registered Medical Practitioner or a Hospital only.

MEGAPTH-G

Teriparatide solution for injection

20 µg / 80 µl

Composition

Each ml of Teriparatide (r- Human Parathyroid Hormone) (1-34)

Injection contains:

Teriparatide250 mcg
Glacial acetic acid Ph. Eur.0.410 mg
Sodium acetate (trihydrate) Ph. Eur.0.166 mg
Mannitol Ph. Eur.45.400 mg
Metacresol Ph. Eur. (preservative)3.000 mg
Water for injections Ph. Eur.q.s
Hydrochloric acid solution Ph. Eur. (for pH adjustment)q.s
Sodium hydroxide solution Ph. Eur. (for pH adjustment) q.s

Description

MEGAPTH-G is a biosimilar medicinal product to reference product, Forteo/Forsteo 250 UG Solution For Injection.

Each dose of 80 microliters contains 20 micrograms of teriparatide*.

One cartridge of 2.4 mL of solution contains 600 micrograms of teriparatide (corresponding to 250 micrograms per mL).

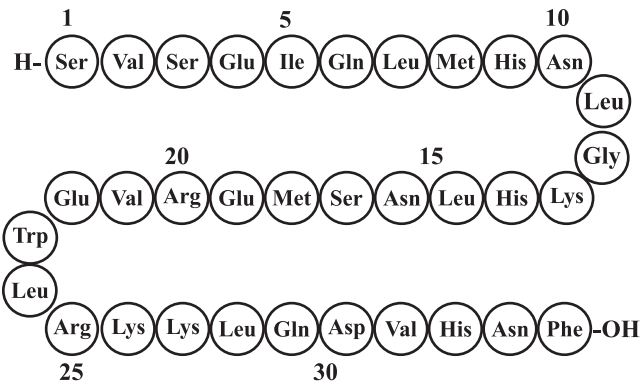
*Teriparatide, rhPTH(1-34), produced in E. coli, using recombinant DNA technology, is identical to the 34-N-terminal amino acid sequence of endogenous human parathyroid hormone.

List of excipients: Glacial acetic acid, Mannitol, Metacresol, Sodium acetate trihydrate, Hydrochloric acid (for pH adjustment), Sodium hydroxide (for pH adjustment), Water for injections

Molecular formula, weight:

Teriparatide: C₁₈₁H₃₀₁N₅₅O₅₁S₂

Teriparatide has a molecular weight of 4117.8 daltons and its amino acid sequence is shown below:



PHARMACEUTICAL FORM

Solution for injection

Clear, colourless solution.

Clinical Pharmacology

Pharmacodynamic properties

Pharmaco-therapeutic group: Calcium homeostasis, parathyroid hormones and analogues, ATC code: H05AA02.

Osteoporosis is characterised by low bone mass and microarchitectural deterioration of bone tissue, leading to bone fragility and an increase in the risk of vertebral and non-vertebral fractures. The diagnosis of osteoporosis may be confirmed by the finding of low bone mass or the presence or history of osteoporotic fracture. While non-vertebral fractures are usually clinically apparent, vertebral fractures also may be manifested by back pain, height loss or kyphosis.

Mechanism of action

Endogenous 84-amino-acid parathyroid hormone (PTH) is the primary regulator of calcium and phosphate metabolism in bone and kidney. Teriparatide is the active fragment (1-34) of endogenous human parathyroid hormone, manufactured using recombinant DNA technology.

Physiological actions of PTH include regulation of bone metabolism, renal tubular reabsorption of calcium and phosphate, and intestinal calcium absorption. The biological actions of PTH and teriparatide are mediated through binding to specific PTH cell surface receptors. Teriparatide binds to these receptors with similar affinity as PTH and has the same actions in bone and kidney as PTH. Like endogenous PTH, teriparatide is not expected to accumulate in bone or other tissues.

Pharmacodynamic effects

The skeletal effects of teriparatide depend upon the pattern of systemic exposure. Once daily administration of teriparatide increases apposition of new bone on trabecular and cortical (endosteal and periosteal) bone surfaces by preferential stimulation of osteoblastic activity over osteoclastic activity. In contrast, continuous excess of endogenous PTH, as occurs in hyperparathyroidism, may be detrimental to the skeleton because bone resorption may be stimulated more than bone formation.

In monkey studies, teriparatide improved trabecular microarchitecture and increased bone mass and strength by stimulating new bone formation in both cancellous and cortical bone. In humans, teriparatide affects calcium and phosphorus metabolism in a pattern consistent with the known actions of endogenous PTH.

Clinical efficacy

Risk factors

Independent risk factors, for example, low BMD, age, the existence of previous fracture, family history of hip fractures, high bone turnover and low body mass index should be considered in order to identify women and men at increased risk of osteoporotic fractures who could benefit from treatment. Premenopausal women with glucocorticoid-induced osteoporosis should be considered at high risk for fracture if they have a prevalent fracture or a combination of risk factors that place them at high risk for fracture (e.g., low bone density [e.g., T-score ≤-2], sustained high dose glucocorticoid therapy [e.g., ≥7.5 mg/day for at least 6 months], high underlying disease activity, low sex steroid levels).

Postmenopausal osteoporosis

The pivotal study included 1,637 postmenopausal women (mean age 69.5 years). At baseline, ninety percent of the patients had one or more vertebral fractures, and on average, vertebral BMD was 0.82 g/cm² (equivalent to a T-score = -2.6). All patients were offered 1,000 mg calcium per day and at least 400 IU vitamin D per day. Results from up to 24 months (median: 19 months) treatment with teriparatide demonstrate statistically significant fracture reduction (Table 1). To prevent one or more new vertebral fractures, 11 women had to be treated for a median of 19 months.

Table 1. Fracture incidence in postmenopausal women

	Placebo (N = 544) (%)	Teriparatide (N = 541) (%)	Relative risk (95% CI) vs. placebo
New vertebral fracture (≥1) ^a	14.3	5.0 ^b	0.35 (0.22, 0.55)
Multiple vertebral fracture (≥2) ^a	4.9	1.1 ^b	0.23 (0.09, 0.60)
Non-vertebral fragility fractures ^c	5.5%	2.6% ^d	0.47 (0.25, 0.87)
Major non-vertebral fragility fractures (hip, radius, humerus, ribs and pelvis)	3.9%	1.5% ^d	0.38 (0.17, 0.86)

Abbreviations: N = number of patients randomly assigned to each treatment group; CI = confidence interval.

a The incidence of vertebral fractures was assessed in 448 placebo and 444 teriparatide patients who had baseline and follow-up spine radiographs.

b p≤0.001 compared with placebo.

c A significant reduction in the incidence of hip fractures has not been demonstrated.

d p≤0.025 compared with placebo.

After 19 months (median) treatment, BMD had increased in the lumbar spine and total hip, respectively, by 9% and 4% compared with placebo (p<0.001).

Post-treatment management: Following treatment with teriparatide, 1,262 postmenopausal women from the pivotal trial enrolled in a post-treatment follow-up study. The primary objective of the study was to collect safety data of teriparatide. During this observational period, other osteoporosis treatments were allowed and additional assessment of vertebral fractures was performed.

During a median of 18 months following discontinuation of teriparatide, there was a 41% reduction (p=0.004) compared with placebo in the number of patients with a minimum of one new vertebral fracture.

In an open-label study, 503 postmenopausal women with severe osteoporosis and a fragility fracture within the previous 3 years (83% had received previous osteoporosis therapy) were treated with teriparatide for up to 24 months. At 24 months, the mean increase from baseline in lumbar spine, total hip and femoral neck BMD was 10.5%, 2.6 % and 3.9% respectively. The mean increase in BMD from 18 to 24 months was 1.4%, 1.2%, and 1.6% at the lumbar spine, total hip and femoral neck, respectively.

A 24-month, randomised, double-blind, comparator-controlled Phase 4 study included 1,360 postmenopausal women with established osteoporosis. 680 subjects were randomised to teriparatide and 680 subjects were randomised to oral risedronate 35 mg/week. At baseline, the women had a mean age of 72.1 years and a median of 2 prevalent vertebral fractures; 57.9% of patients had received previous bisphosphonate therapy and 18.8% took concomitant glucocorticoids during the study. 1,013 (74.5%) patients completed the 24-month follow-up. The mean (median) cumulative dose of glucocorticoid was 474.3 (66.2) mg in the teriparatide arm and 898.0 (100.0) mg in the risedronate arm. The mean (median) vitamin D intake for the teriparatide arm was 1,433 IU/day (1 400 IU/day) and for the

risedronate arm was 1,191 IU/day (900 IU/day). For those subjects who had baseline and follow-up spine radiographs, the incidence of new vertebral fractures was 28/516 (5.4%) in teriparatide- and 64/533 (12.0%) in risedronate-treated patients, relative risk (95% CI) = 0.44 (0.29-0.68), p<0.0001. The cumulative incidence of pooled clinical fractures (clinical vertebral and non-vertebral fractures) was 4.8% in teriparatide and 9.8% in risedronate-treated patients, hazard ratio (95% CI) = 0.48 (0.32-0.74), p=0.0009.

Male osteoporosis

437 patients (mean age 58.7 years) were enrolled in a clinical trial for men with hypogonadal (defined as low morning free testosterone or an elevated FSH or LH) or idiopathic osteoporosis. Baseline spinal and femoral neck BMD mean T-scores were -2.2 and -2.1, respectively. At baseline, 35% of patients had a vertebral fracture and 59% had a non-vertebral fracture.

All patients were offered 1,000 mg calcium per day and at least 400 IU vitamin D per day. Lumbar spine BMD significantly increased by 3 months. After 12 months, BMD had increased in the lumbar spine and total hip by 5% and 1%, respectively, compared with placebo. However, no significant effect on fracture rates was demonstrated.

Glucocorticoid-induced osteoporosis

The efficacy of teriparatide in men and women (N=428) receiving sustained systemic glucocorticoid therapy (equivalent to 5 mg or greater of prednisone for at least 3 months) was demonstrated in the 18-month primary phase of a 36-month, randomised, double-blind, comparator-controlled study (alendronate 10 mg/day). Twenty-eight percent of patients had one or more radiographic vertebral fractures at baseline. All patients were offered 1,000 mg calcium per day and 800 IU vitamin D per day.

This study included postmenopausal women (N=277), premenopausal women (N=67), and men (N=83). At baseline, the postmenopausal women had a mean age of 61 years, mean lumbar spine BMD T score of -2.7, median prednisone equivalent dose of 7.5 mg/day, and 34% had one or more radiographic vertebral fractures; premenopausal women had a mean age of 37 years, mean lumbar spine BMD T score of -2.5, median prednisone equivalent dose of 10 mg/day, and 9% had one or more radiographic vertebral fractures; and men had a mean age of 57 years, mean lumbar spine BMD T score of -2.2, median prednisone equivalent dose of 10 mg/day, and 24% had one or more radiographic vertebral fractures.

Sixty-nine percent of patients completed the 18-month primary phase. At the 18-month endpoint, teriparatide significantly increased lumbar spine BMD (7.2%) compared with alendronate (3.4%) (p<0.001). Teriparatide increased BMD at the total hip (3.6%) compared with alendronate (2.2%) (p<0.01), as well as at the femoral neck (3.7%) compared with alendronate (2.1%) (p<0.05). In patients treated with teriparatide, lumbar spine, total hip and femoral neck BMD increased between 18 and 24 months by an additional 1.7%, 0.9%, and 0.4%, respectively.

At 36 months, analysis of spinal X-rays from 169 alendronate patients and 173 teriparatide patients showed that 13 patients in the alendronate group (7.7%) had experienced a new vertebral fracture compared with 3 patients in the teriparatide group (1.7%) (p=0.01). In addition, 15 of 214 patients in the alendronate group (7.0%) had experienced a non-vertebral fracture compared with 16 of 214 patients in the teriparatide group (7.5%) (p=0.84).

In premenopausal women, the increase in BMD from baseline to 18 month endpoint was significantly greater in the teriparatide group compared with the alendronate group at the lumbar spine (4.2% versus -1.9%; p<0.001) and total hip (3.8% versus 0.9%; p=0.005). However, no significant effect on fracture rates was demonstrated.

Comparability of MEGAPTH-G with FORTEO

Study RGB1023O31

Therapeutic equivalence of MEGAPTH-G and FORSTEO was demonstrated in a Japanese, multicentre, randomised, non-inferiority, active drug controlled, rater-blinded, parallel-group comparative phase III study in men and postmenopausal women with osteoporosis at high risk of fracture (a total of 250 subjects). Subjects were administered MEGAPTH-G or FORSTEO (teriparatide 20 µg/80 µg solution for injection) daily via s.c. injection for 52 weeks. The percent change (mean ± standard deviation [SD]) in lumbar spine (L2-L4) BMD at Week 52, which served as the primary endpoint of the study, was 8.94% ± 6.19% in the MEGAPTH-G group and 9.65% ± 6.22% in the Forteo group. The (adjusted) difference in the mean between the MEGAPTH-G and FORTEO groups was -0.65% (2-sided 95% CI: -2.17% to 0.87%). The 2-sided 95% CI was within the range of the pre-specified equivalence margin (±2.8). There were no substantial differences between the MEGAPTH-G and the Forteo groups in terms of the secondary endpoints of the absolute change in lumbar spine (L2-L4) BMD, the percent and absolute changes in lumbar spine (L1-L4) BMD, femoral neck BMD and proximal femoral BMD, the percent and absolute changes in bone metabolic marker of P1NP and the incidence of vertebral or non-vertebral fractures.

Pharmacokinetics properties

Distribution

The volume of distribution is approximately 1.7 L/kg. The half-life of teriparatide is approximately 1 hour when administered subcutaneously, which reflects the time required for absorption from the injection site.

Biotransformation

No metabolism or excretion studies have been performed with teriparatide but the peripheral metabolism of parathyroid hormone is believed to occur predominantly in liver and kidney.

Elimination

Teriparatide is eliminated through hepatic and extra-hepatic clearance (approximately 62 L/hr in women and 94 L/hr in men).

Elderly

No differences in teriparatide pharmacokinetics were detected with regard to age (range 31 to 85 years). Dose adjustment based on age is not required.

INDICATIONS

MEGAPTH-G is indicated in adults.

Treatment of osteoporosis in postmenopausal women and in men at increased risk of fracture. In postmenopausal women, a significant reduction in the incidence of vertebral and non-vertebral fractures but not hip fractures has been demonstrated.

Treatment of osteoporosis associated with sustained systemic glucocorticoid therapy in women and men at increased risk for fracture.

DOSAGE AND ADMINISTRATION

The recommended dose of MEGAPTH-G is 20 micrograms administered once daily.

The maximum total duration of treatment with MEGAPTH-G should be 24 months. The 24- month course of MEGAPTH-G should not be repeated over a patient's lifetime.

Patients should receive supplemental calcium and vitamin D supplements if dietary intake is inadequate.

Following cessation of MEGAPTH-G therapy, patients may be continued on other osteoporosis therapies.

Route of administration

Subcutaneous (SC)

Method of administration

MEGAPTH-G should be administered once daily by subcutaneous injection in the thigh or abdomen. Patients must be trained to use the proper injection techniques. For instructions of the medicinal product, before administration, see section Special precautions for disposal and other handling and Instructions for use (IFU). Instructions for use of Terrosa Pen, which are provided with the pen is also available to instruct patients on the correct use of the pen.

SPECIAL POPULATIONS

Patients with renal impairment

MEGAPTH-G must not be used in patients with severe renal impairment (see section Contraindications). In patients with moderate renal impairment, MEGAPTH-G should be used with caution. No special caution is required for patients with mild renal impairment.

Patients with hepatic impairment

No data are available in patients with impaired hepatic function (see section Preclinical safety data). Therefore, MEGAPTH-G should be used with caution.

Paediatric population and young adults with open epiphyses

The safety and efficacy of MEGAPTH-G in children and adolescents less than 18 years has not been established. MEGAPTH-G should not be used in paediatric patients (less than 18 years), or young adults with open epiphyses.

Elderly patients

Dosage adjustment based on age is not required (see section pharmacokinetic properties).

Women of childbearing potential/ Contraception in females

Women of childbearing potential should use effective methods of contraception during use of Teriparatide. If pregnancy occurs, Teriparatide should be discontinued.

Pregnancy

Teriparatide is contraindicated for use during pregnancy.

Breast-feeding

Teriparatide is contraindicated for use during breast-feeding. It is not known whether teriparatide is excreted in human milk.

Fertility

Studies in rabbits have shown reproductive toxicity (see section Preclinical safety data). The effect of teriparatide on human foetal development has not been studied. The potential risk for humans is unknown.

Effects on ability to drive and use machines

Teriparatide has no or negligible influence on the ability to drive and use machines. Transient, orthostatic hypotension or dizziness was observed in some patients. These patients should refrain from driving or the use of machines until symptoms have subsided.

Contraindications

- Hypersensitivity to the active substance or to any of the excipients.
- Pregnancy and breast-feeding
- Pre-existing hypercalcaemia
- Severe renal impairment
- Metabolic bone diseases (including hyperparathyroidism and Paget's disease of the bone) other than primary osteoporosis or glucocorticoid-induced osteoporosis.

size

259 × 480 (±0,5)/40 × 70 (±1) mm

- Unexplained elevations of alkaline phosphatase
- Prior external beam or implant radiation therapy to the skeleton
- Patients with skeletal malignancies or bone metastases should be excluded from treatment with teriparatide.

WARNINGS & PRECAUTIONS

Traceability

In order to improve the traceability of biological medicinal products, the name and the batch number of the administered product should be clearly recorded.

Serum and urine calcium

In normocalcaemic patients, slight and transient elevations of serum calcium concentrations have been observed following teriparatide injection. Serum calcium concentrations reach a maximum between 4 and 6 hours and return to baseline by 16 to 24 hours after each dose of teriparatide. Therefore, if blood samples for serum calcium measurements are taken, this should be done at least 16 hours after the most recent Teriparatide injection. Routine calcium monitoring during therapy is not required. Teriparatide may cause small increases in urinary calcium excretion, but the incidence of hypercalciuria did not differ from that in the placebo-treated patients in clinical trials.

Urolithiasis

Teriparatide has not been studied in patients with active urolithiasis. Teriparatide should be used with caution in patients with active or recent urolithiasis because of the potential to exacerbate this condition

Orthostatic hypotension

With the use of Teriparatide, isolated episodes of transient orthostatic hypotension were observed. Typically, an event began within 4 hours of dosing and spontaneously resolved within a few minutes to a few hours. When transient orthostatic hypotension occurred, it happened within the first several doses, was relieved by placing subjects in a reclining position, and did not preclude continued treatment.

Renal impairment

Caution should be exercised in patients with moderate renal impairment.

Younger adult population

Experience in the younger adult population, including premenopausal women, is limited. Treatment should only be initiated if the benefit clearly outweighs risks in this population. Women of childbearing potential should use effective methods of contraception during use of Teriparatide. If pregnancy occurs, Teriparatide should be discontinued.

Duration of treatment

Studies in rats indicate an increased incidence of osteosarcoma with long-term administration of teriparatide (see section Preclinical safety data). Until further clinical data become available, the recommended treatment time of 24 months should not be exceeded.

Sodium content

This medicine contains less than 1 mmol sodium (23 mg) per dose, that is to say essentially 'sodium-free'

Drug Interactions

In a study of 15 healthy subjects administered digoxin daily to steady state, a single Teriparatide dose did not alter the cardiac effect of digoxin. However, sporadic case reports have suggested that hypercalcaemia may predispose patients to digitalis toxicity. Because Teriparatide transiently increases serum calcium, Teriparatide should be used with caution in patients taking digitalis.

Teriparatide has been evaluated in pharmacodynamic interaction studies with hydrochlorothiazide. No clinically significant interactions were noted.

Co-administration of raloxifene or hormone replacement therapy with Teriparatide did not alter the effects of Teriparatide on serum or urine calcium or on clinical adverse events.

ADVERSE REACTIONS

Summary of the safety profile

The most commonly reported adverse reactions in patients treated with Teriparatide are nausea, pain in limb, headache and dizziness.

List of adverse reactions

The adverse reactions associated with the use of teriparatide in osteoporosis clinical trials and postmarketing exposure are summarised in the table below. The following convention has been used for the classification of the adverse reactions: very common (≥1/10), common (≥1/100 to <1/10), uncommon (≥1/1,000 to <1/100), rare (≥1/10,000 to <1/1,000) very rare (<1/10,000).

Blood and lymphatic system disorders

Common: Anaemia

Immune System Disorder

Rare: Anaphylaxis

Metabolism and nutrition disorders

Common: Hypercholesterolaemia

Uncommon: Hypercalcaemia greater than 2.76 mmol/L, hyperuricemia

Rare: Hypercalcaemia greater than 3.25 mmol/L

Psychiatric disorders

Common: Depression

Nervous system disorders

Common: Dizziness, headache, sciatica, syncope

Ear and labyrinth disorders

Common: Vertigo

Cardiac disorders

Common: Palpitations

Uncommon: Tachycardia

Vascular disorders

Common: Hypotension

Respiratory, thoracic and mediastinal disorders

Common: Dyspnoea

Uncommon: Emphysema

Gastrointestinal disorders

Common: Nausea, vomiting, hiatus hernia, gastroesophageal reflux disease

Uncommon: Haemorrhoids

Skin and subcutaneous tissue disorders

Common: Sweating increased

Musculoskeletal and connective tissue disorders

Very common: Pain in limb

Common: Muscle cramps

Uncommon: Myalgia, arthralgia, back cramp/pain*

Renal and urinary disorders

Uncommon: Urinary incontinence, polyuria, micturition urgency, nephrolithiasis

Rare: Renal failure/impairment

General disorders and administration site conditions

Common: Fatigue, chest pain, asthenia, mild and transient injection site events, including pain, swelling, erythema, localised bruising, pruritis and minor bleeding at injection site.

Uncommon: Injection site erythema, injection site reaction

Rare: Possible allergic events soon after injection: acute dyspnoea, oro/facial oedema, generalised urticaria, chest pain, oedema (mainly peripheral).

Investigations

Uncommon: Weight increased, cardiac murmur, alkaline phosphatase increase

*Serious cases of back cramp or pain have been reported within minutes of the injection.

Description of selected adverse reactions

In clinical trials the following reactions were reported at a ≥ 1% difference in frequency from placebo: vertigo, nausea, pain in limb, dizziness, depression, dyspnoea.

Teriparatide increases serum uric acid concentrations. In clinical trials, 2.8% of Teriparatide patients had serum uric acid concentrations above the upper limit of normal compared with 0.7% of placebo patients.

However, the hyperuricemia did not result in an increase in gout, arthralgia, or urolithiasis.

In a large clinical trial, antibodies that cross-reacted with teriparatide were detected in 2.8% of women receiving Teriparatide. Generally, antibodies were first detected following 12 months of treatment and diminished after withdrawal of therapy. There was no evidence of hypersensitivity reactions, allergic reactions, effects on serum calcium, or effects on Bone Mineral Density (BMD) response.

Preclinical Safety Data

A study was performed to determine and compare the potential toxicity of MEGAPTH-G (20 µg of teriparatide) and Forsteo group (Teriparatide 20 µg) when given by daily subcutaneous injection for 4 weeks to Sprague-Dawley rats, and to provide data to support the use of MEGAPTH-G (20 µg of teriparatide) in humans. Administration of Teriparatide or MEGAPTH-G (20 µg of teriparatide) by once daily subcutaneous injection in Sprague-Dawley rats at doses of 30, 100 and 300 µg/kg/day was well tolerated. Comparable increases in bone mass and/or bone geometry were observed for animals treated with Teriparatide or MEGAPTH-G (20 µg of teriparatide) consistent with microscopic changes in multiple bones interpreted to be due to the pharmacological actional of the compound (macroscopic bone

thickening and firm bone marrow, and microscopic increased bone/bone formation, physéal thickening/thinning, and/or focal resorption). Slight changes in hematology and clinical chemistry parameters considered secondary to the pharmacological effects of both test compounds as bone anabolic agents (PTH(1-34)) were associated with treatment related changes in the spleen (increased spleen weights, macroscopic enlargement and increased extramedullary hematopoiesis), and adrenal gland (cortical vacuolation). Macroscopic and microscopic observations at the injection site were generally minor, similar in character and incidence across all groups including the vehicle control, positive control, and treated animals and were therefore considered to be related to the experimental procedure and/or vehicle. In conclusion, administration of Teriparatide or MEGAPTH-G (20 µg of teriparatide) by once daily subcutaneous injection in Sprague-Dawley rats at doses of 30, 100 and 300 µg/kg/day was well tolerated. Exposure to MEGAPTH-G (20 µg of teriparatide) at 100 µg/kg/day and cannot be drawn because of the large variability in exposure levels (high inter-animal variability, high standard deviation). For the low dose groups (30 µg/kg/day) exposure to Teriparatide or MEGAPTH-G (20 µg of teriparatide) tended to be similar.

Teriparatide was not genotoxic in a standard battery of tests. It produced no teratogenic effects in rats, mice or rabbits. There were no important effects observed in pregnant rats or mice administered teriparatide at daily doses of 30 to 1000 µg/kg. However, fetal resorption and reduced litter size occurred in pregnant rabbits administered daily doses of 3 to 100 µg/kg. The embryotoxicity observed in rabbits may be related to their much greater sensitivity to the effects of PTH on blood ionised calcium compared with rodents.

Rats treated with near-life time daily injections had dose-dependent exaggerated bone formation and increased incidence of osteosarcoma most probably due to an epigenetic mechanism. Teriparatide did not increase the incidence of any other type of neoplasia in rats. Due to the differences in bone physiology in rats and humans, the clinical relevance of these findings is probably minor. No bone tumours were observed in ovariectomised monkeys treated for 18 months or during a 3-year follow-up period after treatment cessation. In addition, no osteosarcomas have been observed in clinical trials or during the post treatment follow-up study. Animal studies have shown that severely reduced hepatic blood flow decreases exposure of PTH to the principal cleavage system (Kupffer cells) and consequently clearance of PTH(1-84).

OVERDOSE

Signs and symptoms

Teriparatide has been administered in single doses of up to 100 micrograms and in repeated doses of up to 60 micrograms/day for 6 weeks.

The effects of overdose that might be expected include delayed hypercalcaemia and risk of orthostatic hypotension. Nausea, vomiting, dizziness, and headache can also occur.

Overdose experience based on post-marketing spontaneous reports

In post-marketing spontaneous reports, there have been cases of medication error where the entire contents (up to 800 mcg) of the teriparatide pen have been administered as a single dose. Transient events reported have included nausea, weakness/lethargy and hypotension. In some cases, no adverse events occurred as a result of the overdose. No fatalities associated with overdose have been reported.

Overdose management

There is no specific antidote for Teriparatide. Treatment of suspected overdose should include transitory discontinuation of Teriparatide, monitoring of serum calcium, and implementation of appropriate supportive measures, such as hydration.

Presentation:

MEGAPTH-G is available as 3 mL cartridge (siliconised Type I glass), with a plunger stopper and disc seal (aluminium and rubber liner seals), packed in a plastic tray sealed with lid foil and packed in a carton. Each cartridge contains 2.4 mL of solution corresponding to 28 doses of 20 micrograms (per 80 microliters).

Pack sizes:

MEGAPTH-G 1 cartridge or 3 cartridges.

MEGAPTH-G cartridge and pen pack:

1 inner carton of MEGAPTH-G cartridge (containing 1 cartridge) and 1 inner carton of Terrosa Pen (containing 1 pen). Not all pack sizes may be marketed.

Incompatibilities:

In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.

Storage :

Store in a refrigerator (2 °C – 8 °C). After insertion of the cartridge into the pen, the combined pen and cartridge should be returned to the refrigerator immediately after use.

Do not freeze. Keep the cartridge in the outer carton in order to protect from light.

Do not store the injection device with the needle attached. Do not remove the cartridge from the pen after first use.

For storage conditions after first opening of the medicinal product, see section Shelf life.

Keep out of reach of children

Shelf Life:

2 years

Chemical in-use stability has been demonstrated for 28 days at 2 – 8 °C.

From a microbiological point of view, once opened, the product may be stored for a maximum of 28 days within its shelf life at 2 °C to 8 °C.

Other in-use storage times and conditions are the responsibility of the user.

Controlled Medicine/ Ubat Terkawal

Special precautions for disposal and other handling

MEGAPTH-G solution for injection is supplied in a cartridge. MEGAPTH-G cartridges are to be used in Terrosa Pen reusable, multidose pen exclusively. MEGAPTH-G cartridges must not be used with any other pen. The pen and injection needles are not included with this medicinal product. However, for the treatment initiation a cartridge and pen pack should be used containing one carton of MEGAPTH-G cartridge and one carton of Terrosa Pen.

Each cartridge and pen should be used by only one patient. The pen can be used with injection needles developed according to the pen needle ISO standard of a gauge between 29 G and 31 G (diameter 0.25 – 0.33 mm) and a length between 5 mm to 12.7 mm for subcutaneous injection only.

A new sterile pen needle must be used for every injection.

The expiry date on the cartridge label must always be checked before inserting the cartridge into Terrosa Pen. To avoid medication errors, make sure that the date when starting to use a new cartridge is at least 28 days before its expiry date.

The date of first injection should also be written on the outer carton of MEGAPTH-G (see the provided space on the box: {First use:}).

Before using the pen device for the first time, the patient should read and understand the instructions on how to use the pen which are provided with the pen.

After each injection, the pen should be returned to the refrigerator. After the first use, the cartridge should not be removed from the pen during the 28 days of usage.

Each cartridge should be properly disposed of after 28 days of first use, even if it is not completely empty.

MEGAPTH-G solution for injection must not be transferred to a syringe.

Empty cartridges must not be refilled.

MEGAPTH-G should not be used if the solution is cloudy, coloured or contains visible particles. Any unused product or waste material should be disposed of in accordance with local requirements.

Interchangability and automatic substitution:

MEGAPTH-G has been developed as a similar biological medicinal product to FORTEO/FORSTEO and has been shown to have a comparable quality, safety and efficacy profile to FORTEO/FORSTEO. Therefore, interchangeability with the reference product is possible, as long as this is in agreement with the treating physician in consultation with the patient. However, automatic substitution (i.e the practice by which a different product to that specified on the prescription is dispensed to the patient without the prior informed consent of the treating physician) and active substance-based prescription cannot apply to biologicals, including biosimilars.



Manufactured for:
MEGA LIFESCIENCES (AUSTRALIA) PTY LTD
Victoria 3810, Australia.

Product Registration Holder (in Malaysia):
MEGA LIFESCIENCES SDN BHD
B-28-02, The Ascent, Paradigm.
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Selangor, Malaysia.

Manufactured by :
GEDEON RICHTER PLC.
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Released by:
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Gyömrői út 19-21, 1103 Budapest, Hungary.

Date of Revision : September 2023

K-33124-1.0

size

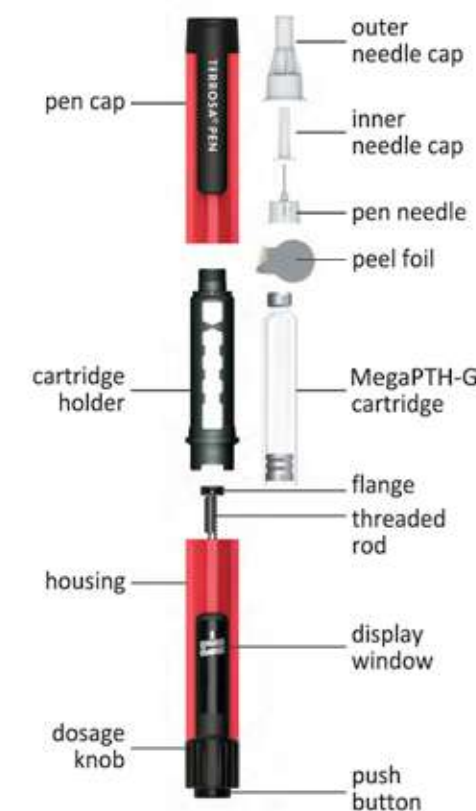
259 × 480 (±0,5)/40 × 70 (±1) mm

■ Black

Re-usable pen injector to be used with MegaPTH-G cartridges for subcutaneous injections

Always follow instructions provided below and on the back page when using the Terrosa Pen.

Terrosa Pen parts



K-33128-1.0, 00000000

Pen preparation - First use/ changing cartridges

Follow the instructions every time you insert a new MegaPTH-G cartridge into your Terrosa Pen. Do not repeat this before each daily injection, otherwise you will not have enough Terrosa for 28 days. Read the patient information leaflet for MegaPTH-G cartridge provided separately.

A: Remove the pen cap.



B: Remove the cartridge holder by turning it (bayonet coupling).



C: Remove the empty cartridge, in the case of changing cartridge. Insert a new MegaPTH-G cartridge into the cartridge holder, with the metal crimped cap of the cartridge first.



Write down the first injection date of each new cartridge. This helps you to know when the 28 daily doses per cartridge are used.

D: Push back the threaded rod carefully with your finger, in a straight line and as far as it will go. This is not necessary when the rod is already in the starting position, such as at the very first use. The threaded rod cannot be pushed back to the pen housing entirely.



E: Attach the cartridge holder to the housing by turning it 90 degrees, until it stops.



F: Attach a new pen needle as follows:



Screw the pen needle clockwise onto the cartridge holder. Make sure that the pen needle is attached correctly and sits firmly on the cartridge holder.



Remove the outer needle cap and save it.



Remove and dispose the inner needle cap.



While attaching the needle some drops may escape, this is normal.

G: Priming
The pen must be primed and tested after inserting a new cartridge and prior the first injection from each cartridge.

Turn the dose knob clockwise until you see a droplet sign on the dose display. Make sure that the two indicator stripes are aligned. During dose dialling the pen provides an audible clicking sound and a noticeable resistance.



Hold the pen with the needle pointing upwards.
Press the push button fully in. Hold it pressed in until the dose indication has turned back to the start position. Some drops of medicine must be expelled from the needle tip.

If no drops emerge, repeat step G until you see some drops. Do not repeat step G more than four times, but follow the instructions given under Troubleshooting section on the back page.



Administration using the Terrosa Pen

Wash your hands carefully with soap to minimise the risk of infection. Make sure you have ready:

- your Terrosa Pen with inserted cartridge
- a compatible pen needle
- a puncture-resistant sharps disposal container for used needles.

Do not use the pen if the cartridge is cloudy, discoloured or contains particles. Read the patient information leaflet for Terrosa cartridge provided separately.

1. Attach the pen needle

Use a new needle for every injection. Do not use the pen needle if the packaging is damaged or not opened by yourself.

Note: There is no need to change the needle when using directly after the pen preparation. In this case continue with step "2. Setting the dose and injection".

Pull off the peel foil.



Screw the pen needle clockwise onto the cartridge holder. Make sure that the pen needle is attached correctly and sits firmly on the cartridge holder.



Remove the outer needle cap and save it.



Remove and dispose the inner needle cap.



While attaching the needle some drops may escape, this is normal.

2. Setting the dose and injection

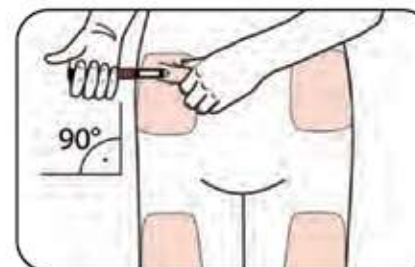
Warning: Ensure the use of the correct drug liquid. Check the label of the cartridge before its insertion into the cartridge holder.

To set the fixed daily dose of 80 microliters, turn the dose knob clockwise until it stops and cannot be rotated any further. Make sure the display shows an arrow sign and it is aligned with the indicator stripe. During dose dialling the pen provides an audible clicking sound and a noticeable resistance. Do not try to force the dose knob any further.



Note: If the cartridge contains less than 80 microliters, the dose knob cannot be turned clockwise up to the arrow sign. In this case, remove the pen needle, change the cartridge and proceed with priming according the pen preparation steps.

Select an appropriate injection site and prepare your skin as recommended by your doctor. Gently hold a skin fold between the thumb and index finger. Insert the needle straight and gently into the skin, as shown in the illustration.



Warning: Prevent the pen needle from bending or breaking. Do not tilt the pen after the needle has been inserted into the skin. Tilting the pen may result in bending or breaking of the needle. Broken needles can get stuck in the skin. Immediately consult the doctor if a broken needle gets stuck in the skin.

Press the push button until the dose indication has returned to the start position. Keep the needle in the skin fold for a further 6 seconds.

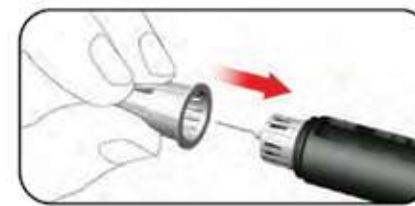


Pull out the pen slowly. Check if the display is at the start position to make sure that the full dose has been injected.



3. Removing the pen needle

Carefully attach the outer needle cap on the pen needle.



Screw the needle cap anti-clockwise to remove the pen needle. Dispose of it correctly, for example, in a puncture-resistant sharps disposal container.



4. Re-attach pen cap

Do not remove the cartridge from the Terrosa Pen before it is empty. Re-attach the pen cap after each use.



size

780 x 176/65 x 176 mm

technical drawing

PIL_Z_Fold_65x176_780x176_temp




Manufacturer:
Ypsomed AG
Brunnmattstrasse 6
3401 Burgdorf
Switzerland

This instruction for use was last revised:
May 2021.

- Put the Terrosa Pen, with the cartridge inserted, back into the refrigerator between 2 and 8 °C immediately after use.

Note for healthcare professionals
Local, healthcare professional or institutional policies may replace the instructions regarding needle handling and disposal.

Additional information

The reusable fixed dose pen is designed for easy administration of Terrosa to treat osteoporosis. Each MegaPTH-G cartridge contains 28 doses of fixed 80 microlitres Terrosa.

Use your Terrosa Pen only as prescribed by your doctor, in this instructions for use and the Terrosa package leaflet.

The Terrosa Pen can be used by self-injecting patients above the age of 18 years, healthcare professionals or third parties such as, for instance, adult relatives.

The Terrosa Pen must not be used by blind or visually impaired patients without help from a trained able-bodied person. Consult your doctor in the case of hearing or handling problems.

If you have any questions concerning the use of the Terrosa Pen, contact our **customer service** at any time.
telephone number: 0044 (0) 207 604 8800
e-mail: medinfo.uk@gedeonrichter.eu

Compatible pen needles

- Ypsomed mylife™ Clickfine® 29 to 31 gauge (diameter 0.25 – 0.33 mm) and 12, 10, 8 or 6 mm length
- BD Micro-Fine Ultra™ needles 29 to 31 gauge (diameter 0.25 – 0.33 mm) and 12.7, 8 or 5 mm length

Pen needles from other manufacturers can be used according to their stated compatibility details.

The **pen needles must be used only once** and the MegaPTH-G cartridge must be used by one person only.

Storage and care of Terrosa Pen

- Handle your pen with care. Do not drop your pen and avoid knocking it against hard surfaces. Protect it from water, dust and moisture.
- A damp cloth is sufficient to clean the Terrosa Pen. Do not use alcohol, other solvents or cleaning agents. Never immerse the Terrosa Pen in water, as this could damage the pen.
- Do not use your Terrosa Pen if it is damaged or if you have any doubts about its correct functioning.
- Transport and store the Terrosa Pen with inserted cartridge at temperatures specified in the Terrosa package leaflet, provided separately.
- Store your Terrosa Pen, cartridges and pen needles out of reach of children.
- Do not store the Terrosa Pen with a needle attached as this may cause air bubbles to form in the cartridge.

Disposal of Terrosa Pen and used accessories

The Terrosa Pen has a service life of two years. Before disposing of the Terrosa Pen always remove the pen needle and the cartridge. Needles and used cartridges must be disposed of separately and safely. The Terrosa Pen can be disposed of according to the instructions of the local authorities.

Warnings

Follow the instructions presented in this instructions for use. If the instructions are not followed, there are risks of incorrect medication, inaccurate dosage, disease

transmission or infection. Seek immediate medical advice if you have any health concern.

Warranty

The warranty covers manufacturing and material defects of your Terrosa Pen for two years of use, from purchase. It is limited to replacement of the pen. The warranty does not cover damages caused by:

- using cartridges other than MegaPTH-G cartridges
- improper or careless use, handling or cleaning
- use contrary to the instructions for use
- the pen being used with a medical device, accessories or consumables other than those mentioned in this Instructions for use
- dropping, impact, application of force, contact with fluids
- other cases of exposure and wear, not in accordance with the instructions for use.

Troubleshooting

Follow the instructions given in the table if you have any questions regarding the use of Terrosa Pen:

Question	Answer
1. Small air bubbles are visible in the cartridge.	A small air bubble will not affect the dose, or harm.
2. Needle cannot be attached.	Use another needle instead. Contact customer service if the second needle cannot be attached.
3. Needle is broken/curved/kinked.	Use another needle instead.
4. During dose dialling the pen does not provide an audible signal.	Do not use this pen; contact customer service.
5. No medicine comes out of the needle during pen preparation step "G: Priming".	Change the needle and repeat the priming as described in pen preparation sections "F" and "G". If still no medicine is expelled, do not use this pen; contact customer service.
6. The dose knob cannot be turned clockwise up to the arrow sign.	The amount of Terrosa left in the cartridge is less than 80 microlitres. Change the cartridge and the pen needle and perform priming according to the pen preparation.
7. Display does not return to start position after injection.	Do not repeat the injection on the same day. Use a new needle for your injection on the following day. Set the dose and complete the injection as described in section "2. Setting the dose and injection". If the display still does not return to start position after injection, do not use this pen; contact customer service.
8. Spillage from pen is observed.	Do not use this pen; contact customer service.
9. The dose knob inadvertently turned clockwise after completing the injection. How do I reset the dose knob to the start position?	Do not press the push button. Reset the pen by simply turning back the dose knob anti-clockwise to the start position.

Instructions for use Terrosa® Pen

